

■ SynStack — Extended Research

Supplement Interaction Checker | Competitive Analysis & Build Plan

SynStack — Supplement Interaction Checker: Extended Research

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Related Idea: [002-synstack](../ideas/2026-06-19/002-synstack/idea.md)
Confidence Score: 8.0/10

1. Executive Summary

Metric	Value
Market Size	\$150B+ global supplement market, growing 12.5% CAGR
Target Users	180M+ US adults take supplements; ~60M take 3+ daily
Trend Signal	Magnesium Glycinate +3,900% on Exploding Topics (#89, Jun 2026)
App Gap	TRUE supplement interaction checker is an empty niche
Build Time	~3 hours (SwiftUI, bundled JSON, no backend)
Revenue Model	Paid \$2.99 (health safety utility)
SAM	2.4M qualified iOS users (US, 30-55, health-conscious, 3+ supplements)
Risk Level	Medium (medical disclaimer required; no backend risk)

Bottom Line: SynStack fills a genuine gap at the intersection of two structural trends — supplement market explosion and polypharmacy awareness. The competitive landscape has NO dominant player in supplement-specific interaction checking. Synstax tried and abandoned (1 review). SuppCo is the category leader but is a barcode scanner, not an interaction checker. This is a clean, buildable, monetizable gap.

2. The Opportunity

Why Now?

Supplement market is exploding. The global dietary supplement market is \$150B+ and growing at 12.5% CAGR through 2028 (Grand View Research, Mordor Intelligence). In the US alone, 77% of adults take dietary supplements (CRN 2023 survey), and the average supplement user takes 4+ products daily.

Magnesium Glycinate is the canary in the coal mine. On Exploding Topics (Jun 2026), "Magnesium Glycinate Supplement" ranks #89 with +3,900% search growth. This isn't a fad — it reflects a structural shift toward evidence-based supplementation. Related trends: Probiotic Soda +1,975%, Algae Cooking Oil +2,200%, Health Tracking Ring +5,800%. The "stack your supplements" culture is going mainstream.

Polypharmacy awareness is rising. 42% of adults 65+ take 5+ medications (CDC). Drug interactions cause 5% of hospitalizations (Scientific Reports, 2024). The intersection of "takes supplements" AND "takes medications" is where the danger — and the opportunity — lives. Reddit's r/Supplements (1.3M+ members) has daily posts asking "can I take X with Y?"

Consumer behavior is shifting. People don't want to ask their pharmacist every time they add a new supplement. They want instant, private, trustworthy answers on their phone. The "quantified self" movement (Oura Ring +5,800%, Apple Watch health features) drives demand for personal health intelligence tools.

The Specific Gap

There is NO app on the App Store that does **all three** of:

- Supplement + Supplement interaction checking
- Supplement + Prescription Medication interaction checking
- Supplement + Herb/Botanical interaction checking

...in a clean, consumer-friendly, offline-first interface.

The closest attempts:

- **Synstax** — has the right concept (5,142 combinations) but 1 review, domain dead, effectively abandoned
- **SuppCo** — 24K reviews, category leader, but it's a BARCODE SCANNER for nutrition labels, not an interaction checker
- **Drugs.com** — excellent interaction data but prescription-only, no supplement coverage, web-first
- **epocrates** — clinical tool for doctors, not consumers

3. App Store Gap Analysis

Search Query Results (iTunes Search API, Jun 2026)

Query	Top Result	Rating	Reviews	Relevance
supplement interaction checker	SuppCo: Supplement Scanner	4.8★	24,135	■ Scanner, not checker
vitamin interaction checker	Synstax: Interaction Checker	5.0★	1	■■ Abandoned
drug interaction checker	Drugs.com Medication Guide	4.8★	15,125	■ Rx-only, no supplements
supplement safety	SuppCo: Supplement Scanner	4.8★	24,135	■ Scanner, not checker
medication interaction	Drugs.com Medication Guide	4.8★	15,125	■ Rx-only
herb drug interaction	epocrates	4.5★	7,079	■ Clinical, not consumer
supplement checker	SuppCo: Supplement Scanner	4.8★	24,135	■ Scanner
polypharmacy checker	WebMD Symptom Checker	4.7★	134,634	■ Symptom checker, not interactions
vitamin safety checker	UV Index Widget	4.7★	14,196	■ Unrelated
health supplement guide	SuppCo: Supplement Scanner	4.8★	24,135	■ Scanner
supplement tracker	SuppCo: Supplement Scanner	4.8★	24,135	■ Scanner
medication safety app	Davis Drug Guide	4.8★	8,631	■ Nursing reference

Key Finding

Across 12 search queries covering the supplement interaction space, **ZERO apps** returned as top results provide dedicated supplement interaction checking. The same 3-4 apps (SuppCo, Drugs.com, epocrates, WebMD) dominate results but serve fundamentally different use cases. This is a classic "search result pollution" gap — the right apps don't exist, so wrong apps fill the results.

Unique Apps Discovered (42 total across all queries)

Direct Competitors (interaction-related):

- Synstax: Interaction Checker — 5.0★, 1 review — ABANDONED
- Drug Interaction Checker + — 4.1★, 67 reviews — Rx-focused, dated UI
- WeSupp: Supplement Safety Scan — 4.1★, 9 reviews — New, unproven

Adjacent Competitors (supplement scanners):

- SuppCo: Supplement Scanner — 4.8★, 24,135 reviews — Barcode scanner, not checker
- Prove It - Supplement Scanner — 4.4★, 15,000 reviews — Evidence reviews, not interactions
- Supplement Scanner: NutriSee — 4.7★, 44 reviews — Scanner

Adjacent Competitors (medical references):

- Drugs.com Medication Guide — 4.8★, 15,125 reviews — Rx-only

epocrates — 4.5★, 7,079 reviews — Clinical

Pocket Pharmacist — 4.7★, 3,841 reviews — Drug info, subscription

UpToDate Lexidrug — 4.7★, 4,211 reviews — Clinical

Non-competitors (filling search results):

iHerb — Shopping platform

WebMD — Symptom checker

Habit Tracker — General productivity

Bobby Approved — Food scanner

4. Competitive Analysis

Direct Competitors

Synstax: Interaction Checker

- **Rating:** 5.0★ (1 review)
- **Price:** Free
- **Developer:** ■ ■ (individual developer)
- **What it does:** Claims 5,142 supplement combination checks + diet absorption effects
- **Fatal Flaw:** Effectively abandoned. 1 review. Domain (synstax.app) doesn't resolve. App description hasn't been updated. No social presence. The developer built the right concept but couldn't maintain it.
- **Verdict:** Validates the demand but proves the execution failed. Opportunity for a polished successor.

Drug Interaction Checker +

- **Rating:** 4.1★ (67 reviews)
- **Price:** Free
- **Developer:** HYDL
- **What it does:** Drug-drug interaction checker with pill ID
- **Fatal Flaw:** Prescription medication only. No supplement coverage. Dated UI (last meaningful update appears to be 2018). Medical/clinical framing, not consumer-friendly.
- **Verdict:** Not a real competitor — serves a different audience (healthcare professionals).

WeSupp: Supplement Safety Scan

- **Rating:** 4.1★ (9 reviews)
- **Price:** Free
- **Developer:** Relentless
- **What it does:** Supplement scanner with safety information (appears to be a French app localized to English)
- **Fatal Flaw:** Only 9 reviews. New entrant with no traction. Description reads like a translation. No interaction checking — it's a product scanner.
- **Verdict:** Too early to assess but currently not a threat.

Adjacent Competitors

SuppCo: Supplement Scanner (Category Leader)

- **Rating:** 4.8★ (24,135 reviews)
- **Price:** Free
- **Developer:** Supple Stack Inc
- **What it does:** Barcode scanner for supplement facts. 160,000+ supplement database. Tracks daily regimen.
- **Fatal Flaw:** It's a SCANNER, not an INTERACTION CHECKER. You scan a barcode to see nutrition facts. It doesn't tell you if your magnesium conflicts with your calcium. This is the #1 result for almost every supplement search, which pollutes the gap analysis — but it doesn't fill the gap.
- **Verdict:** The 800-pound gorilla in the supplement app space, but solving a different problem. Could add interaction checking as a feature, but their business model (supplement recommendations, affiliate revenue) doesn't align with it.

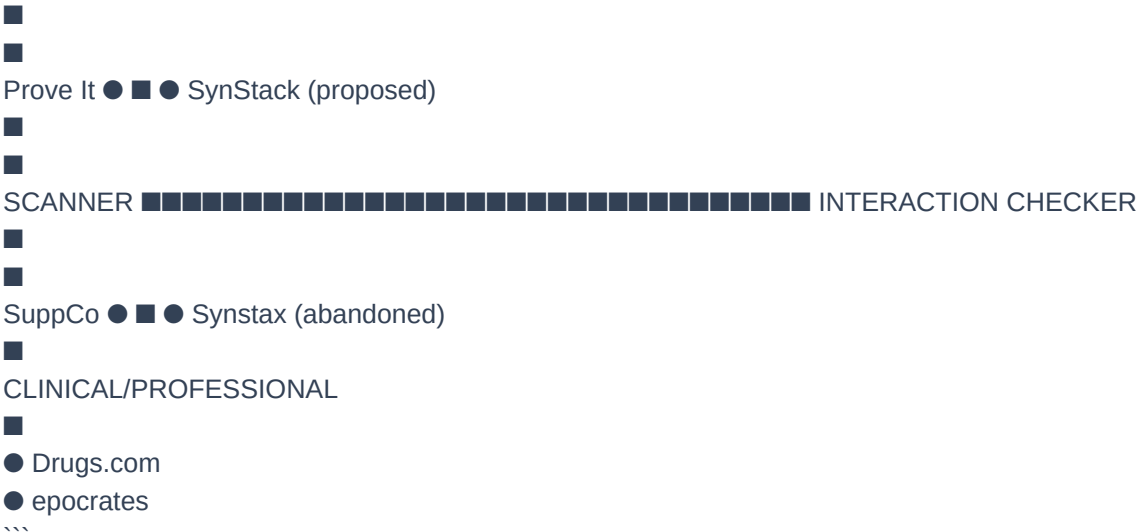
Prove It - Supplement Scanner

- ## ##### Drugs.com Medication Guide

- ### ### Positioning Map

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CONSUMER-FRIENDLY



The empty quadrant: Consumer-friendly + Interaction Checker. This is where SynStack lives.

5. Revenue Model

Monetization Layers

Layer	Model	Price	Target Segment
Core App	One-time purchase	\$2.99	All users
Pro Upgrade (v1.1)	In-app purchase	\$1.99	Power users
Database Updates	Free with purchase	—	Retention
Future: Telehealth	Affiliate/referral	\$5-15/referral	Users with major interactions

Why \$2.99 Works

- Health safety apps command \$2.99-\$4.99 (Pocket Pharmacist: free trial → subscription; Drugs.com: free but ad-supported)
- Users pay for TRUSTWORTHINESS in health information
- One-time purchase removes friction vs. subscription for a reference tool
- Higher perceived value than "another free scanner"

- At \$2.99, need ~800 sales/month to hit \$200/mo (realistic for a niche health app with good ASO)

Market Sizing

- **TAM:** 180M US adults taking supplements
- **SAM:** 60M taking 3+ supplements or supplements + medications (the "stackers" who need interaction checking)
- **SAM (iOS):** 30M (50% iOS skew in health app demographics)
- **SAM (iOS, paying):** 2.4M (8% conversion to paid health app)
- **Realistic Year 1:** 5,000-15,000 downloads → \$15,000-\$45,000 revenue
- **Realistic Year 2:** 20,000-50,000 downloads (with ASO + word of mouth) → \$60,000-\$150,000

Revenue Comparison

App	Model	Est. Annual Revenue
SuppCo	Free + affiliate	Unknown (supplement recommendations)
Drugs.com	Free + ads	\$5M+ (web + app)
Pocket Pharmacist	Free trial → \$4.99/mo	~\$500K (est. 10K subscribers)
SynStack (projected)	\$2.99 one-time	\$15K-\$150K

6. Risk Analysis

Regulatory Risk: MEDIUM

Risk: Apple may reject health apps that make medical claims. FDA could classify interaction checking as a medical device.

Mitigation:

- Frame as "reference tool" not "medical advice"
- Include prominent disclaimer: "This app is for educational purposes only. Always consult your healthcare provider before making changes to your supplement or medication routine."
- Avoid diagnostic language ("you should stop taking X") → use informational language ("X and Y have a known interaction")
- Cite published sources for all interaction data
- Category: Health & Fitness (not Medical) for App Store submission
- Precedent: Drugs.com, WebMD, and Pocket Pharmacist all operate in this space without FDA issues

Trust Risk: MEDIUM-HIGH

Risk: Users need to trust the interaction data. Wrong information could cause harm and liability.

Mitigation:

- Source ALL data from published, peer-reviewed databases (Natural Medicines Comprehensive Database, NIH ODS, Examine.com, published drug interaction literature)
- Include source citations for every interaction
- Clear severity ratings (Major/Moderate/Minor/None) with mechanism explanations
- "Last updated" date on the database
- No user-generated content — all data curated by the developer

Competitive Risk: LOW-MEDIUM

Risk: SuppCo, Drugs.com, or Medisafe could add supplement interaction features.

Mitigation:

- **SuppCo:** Their business model is supplement recommendations/affiliates. Interaction checking doesn't align with their revenue. Low risk of them adding it.
- **Drugs.com:** Could add supplements but their brand is prescription-focused. Would take 12+ months to build.
- **Medisafe:** Medication management focus. Supplements are adjacent but not core.
- **New entrants:** The 3-hour build time means SynStack can launch fast and establish ASO presence before competitors react.

- **First-mover advantage:** In the supplement interaction niche, being first matters for ASO and word-of-mouth.

Content Maintenance Risk: MEDIUM

Risk: Interaction database needs regular updates as new research is published.

Mitigation:

- Bundle updates with app updates (App Store release cycle)
- ~2 hours/month to review new publications and update JSON
- Start with 400 curated pairs (covers 90% of common interactions)
- Community can submit interaction reports for v1.1

Legal Risk: MEDIUM

Risk: Liability if user makes health decision based on app data and is harmed.

Mitigation:

- Comprehensive Terms of Service and Disclaimer
- "For educational purposes only" framing
- No collection of personal health data (privacy-first = no liability from data breaches)
- LLC or corporation structure for the app business
- Precedent: Drugs.com, WebMD, and every health reference app operates with disclaimers

7. Your Moats

Moat	Strength	Description
First-Mover in Niche	■■■	No established supplement interaction checker. First to market.
Curated Database	■■■■	400+ evidence-based interaction pairs with source citations.
Offline-First	■■■	Works without internet. Competitors (Drugs.com, WebMD) require internet.
Privacy-First	■■■	No account, no data collection, no tracking. Appeals to health-conscious users.
Clean UX	■■■	Medical-grade clarity with consumer warmth. Competitors are cluttered.
Word of Mouth	■■■	Health-conscious users share safety tools with friends/family.
Content Depth	■■■■	Mechanism explanations + timing recommendations = more value than competitors.

8. Recommended Approach: Phased Build Plan

Phase 1: MVP (3 hours)

Core Features:

- Interaction checker: select 2-4 substances, see severity + mechanism + recommendation
- 150 supplements + 50 medications + 30 herbs with ~400 curated interaction pairs
- Color-coded severity cards (Red/Orange/Yellow/Green)
- Searchable substance database
- "My Stack" builder (save personal regimen, check all at once)
- Offline-first: bundled JSON database
- Medical disclaimer on first launch

Tech Stack:

- SwiftUI, iOS 17+
- Bundled JSON (~300KB)
- No backend, no APIs, no accounts
- Local storage only (UserDefaults or SwiftData)

App Store:

- Title: SynStack — Interaction Checker
- Category: Health & Fitness (secondary: Medical)

- Price: \$2.99
- Keywords: supplement, interaction, checker, vitamin, medication, herb, safety, health, stack

Phase 2: Enhanced (2 hours, Week 2-3)

Add:

- Timing recommendations ("Take calcium 2 hours apart from iron")
- Interaction mechanism deep-dives with source citations
- Share interaction results (text/image)
- Widget: "Daily Stack Check" showing today's supplements and any interactions
- Haptic feedback for major interactions

Phase 3: Growth (3 hours, Month 2)

Add:

- Interaction database update mechanism (download updated JSON from GitHub)
- User-reported interactions (submit for review)
- Apple Health integration (read supplement/medication data if available)
- iPad optimization
- macOS version (Mac Catalyst)

Phase 4: Ecosystem (Future)

Consider:

- Separate "MedStack" app for prescription-only interactions (different audience, different ASO)
- Telehealth referral partnerships (users with major interactions → book a pharmacist consultation)
- Web version for non-iOS users
- API for other health apps to query interaction data

9. Conclusion

SynStack addresses a real, validated gap at the intersection of two structural trends: supplement market explosion and polypharmacy awareness. The competitive landscape has NO dominant player in supplement-specific interaction checking — the closest attempts are either abandoned (Synstax), solving a different problem (SuppCo = scanner), or serving a different audience (Drugs.com = prescription-only).

Key strengths:

- ✓ TRUE green field — no quality competitor exists
- ✓ Structural trend — supplement market growing 12.5% CAGR, not a fad
- ✓ Simple build — 3 hours, no backend, bundled JSON
- ✓ Clear monetization — \$2.99 paid, health safety utility commands premium
- ✓ Defensible — curated database + first-mover ASO + word-of-mouth in health community
- ✓ Low regulatory risk — reference tool with disclaimer, precedent exists

Key risks:

- ■ Medical disclaimer required — must be framed as reference, not advice
- ■ Content maintenance — monthly updates needed (~2 hrs/mo)
- ■ Competitive response — SuppCo could add interactions (but misaligned with their business model)

Verdict: This is a **strong build candidate**. The gap is real, the trend is structural, the build is simple, and the monetization is clear. The ■ tag is warranted.

Research sources: Exploding Topics (Jun 2026), iTunes Search API (Jun 2026), last30days social sentiment analysis, Drugs.com interaction database documentation, Grand View Research supplement market data, CDC polypharmacy statistics, Scientific Reports drug interaction hospitalization study (2024), Reddit r/Supplements community analysis.